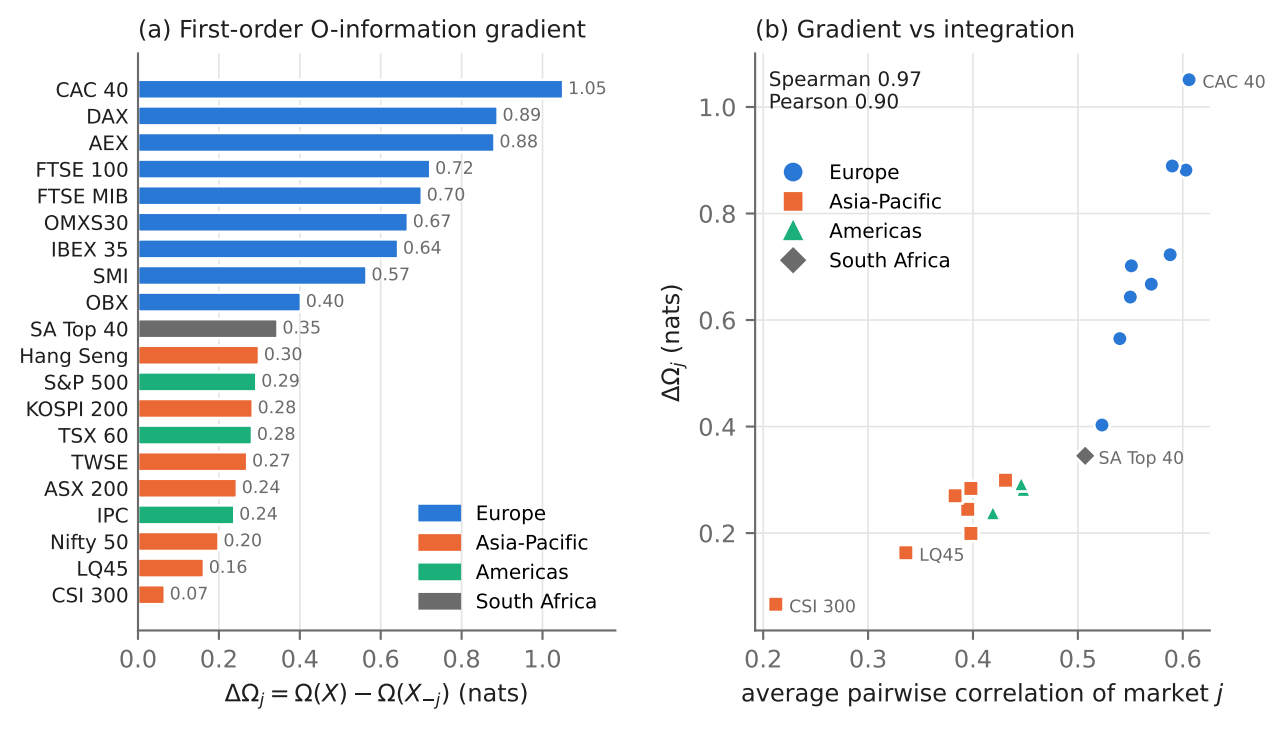
(a) First-order O-information gradient
CAC 40
DAX
AEX
FTSE 100
FTSE MIB
OMXS30
IBEX 35
SMI
OBX
SA Top 40
Hang Seng
S&P 500
KOSPI 200
TSX 60
TWSE
ASX 200
IPC
Nifty 50
LQ45
CSI 300
1.05
0.89
0.88
0.72
0.70
0.67
0.64
0.57
0.40
0.35
0.30
0.29
0.28
0.28
0.27
0.24
0.24
0.20
0.16
0.07
Europe
Asia-Pacific
Americas
South Africa
0.0
0.2
0.4
0.6
0.8
1.0
$\Delta\Omega_j = \Omega(X) - \Omega(X_{-j})$ (nats)
(b) Gradient vs integration
Spearman 0.97
Pearson 0.90
Europe
Asia-Pacific
Americas
South Africa
CAC 40
SA Top 40
LQ45
CSI 300
1.0
0.8
0.6
0.4
0.2
$\Delta\Omega_j$ (nats)
0.2
0.3
0.4
0.5
0.6
average pairwise correlation of market j